

METABRIC Breast Cancer Survival Analysis

2025-06-24

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.3
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
## Warning: package 'readr' was built under R version 4.4.3
```

```
## Warning: package 'purrr' was built under R version 4.4.3
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
## Warning: package 'stringr' was built under R version 4.4.3
```

```
## Warning: package 'forcats' was built under R version 4.4.3
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.0      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr   1.6.0
```

```
## v ggplot2    4.0.2      v tibble    3.3.1
```

```
## v lubridate  1.9.5      v tidyr     1.3.2
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(survival)
```

```
## Warning: package 'survival' was built under R version 4.4.3
```

```
library(survminer)
```

```
## Warning: package 'survminer' was built under R version 4.4.3
```

```
## Loading required package: ggpubr
```

```
## Warning: package 'ggpubr' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'survminer'
```

```
##
```

```
## The following object is masked from 'package:survival':
```

```
##
```

```
## myeloma
```

```
# Importing the dataset for analysis
```

```
metabric <- read_csv("Breast Cancer METABRIC.csv")
```

```
## Rows: 2509 Columns: 34
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (24): Patient ID, Type of Breast Surgery, Cancer Type, Cancer Type Detai...
```

```
## dbl (10): Age at Diagnosis, Cohort, Neoplasm Histologic Grade, Lymph nodes e...
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
colnames(metabric)
```

```
## [1] "Patient ID" "Age at Diagnosis"
## [3] "Type of Breast Surgery" "Cancer Type"
## [5] "Cancer Type Detailed" "Cellularity"
## [7] "Chemotherapy" "Pam50 + Claudin-low subtype"
## [9] "Cohort" "ER status measured by IHC"
## [11] "ER Status" "Neoplasm Histologic Grade"
## [13] "HER2 status measured by SNP6" "HER2 Status"
## [15] "Tumor Other Histologic Subtype" "Hormone Therapy"
## [17] "Inferred Menopausal State" "Integrative Cluster"
## [19] "Primary Tumor Laterality" "Lymph nodes examined positive"
## [21] "Mutation Count" "Nottingham prognostic index"
## [23] "Oncotree Code" "Overall Survival (Months)"
## [25] "Overall Survival Status" "PR Status"
## [27] "Radio Therapy" "Relapse Free Status (Months)"
## [29] "Relapse Free Status" "Sex"
## [31] "3-Gene classifier subtype" "Tumor Size"
## [33] "Tumor Stage" "Patient's Vital Status"
```

```
# Inspecting the structure of our dataset
```

```
str(metabric)
```

```
## spc_tbl_ [2,509 x 34] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
```

```
## $ Patient ID : chr [1:2509] "MB-0000" "MB-0002" "MB-0005" "MB-0006" ...
```

```
## $ Age at Diagnosis : num [1:2509] 75.7 43.2 48.9 47.7 77 ...
```

```
## $ Type of Breast Surgery : chr [1:2509] "Mastectomy" "Breast Conserving" "Mastectomy" "Maste
```

```
## $ Cancer Type : chr [1:2509] "Breast Cancer" "Breast Cancer" "Breast Cancer" "Bre
```

```

## $ Cancer Type Detailed      : chr [1:2509] "Breast Invasive Ductal Carcinoma" "Breast Invasive I
## $ Cellularity               : chr [1:2509] NA "High" "High" "Moderate" ...
## $ Chemotherapy              : chr [1:2509] "No" "No" "Yes" "Yes" ...
## $ Pam50 + Claudin-low subtype : chr [1:2509] "claudin-low" "LumA" "LumB" "LumB" ...
## $ Cohort                   : num [1:2509] 1 1 1 1 1 1 1 1 1 ...
## $ ER status measured by IHC  : chr [1:2509] "Positive" "Positive" "Positive" "Positive" ...
## $ ER Status                 : chr [1:2509] "Positive" "Positive" "Positive" "Positive" ...
## $ Neoplasm Histologic Grade  : num [1:2509] 3 3 2 2 3 3 2 3 2 3 ...
## $ HER2 status measured by SNP6 : chr [1:2509] "Neutral" "Neutral" "Neutral" "Neutral" ...
## $ HER2 Status               : chr [1:2509] "Negative" "Negative" "Negative" "Negative" ...
## $ Tumor Other Histologic Subtype: chr [1:2509] "Ductal/NST" "Ductal/NST" "Ductal/NST" "Mixed" ...
## $ Hormone Therapy           : chr [1:2509] "Yes" "Yes" "Yes" "Yes" ...
## $ Inferred Menopausal State  : chr [1:2509] "Post" "Pre" "Pre" "Pre" ...
## $ Integrative Cluster       : chr [1:2509] "4ER+" "4ER+" "3" "9" ...
## $ Primary Tumor Laterality   : chr [1:2509] "Right" "Right" "Right" "Right" ...
## $ Lymph nodes examined positive : num [1:2509] 10 0 1 3 8 0 1 NA 1 11 ...
## $ Mutation Count            : num [1:2509] NA 2 2 1 2 4 4 NA 1 5 ...
## $ Nottingham prognostic index : num [1:2509] 6.04 4.02 4.03 4.05 6.08 ...
## $ Oncotree Code             : chr [1:2509] "IDC" "IDC" "IDC" "MDLC" ...
## $ Overall Survival (Months)  : num [1:2509] 140.5 84.6 163.7 164.9 41.4 ...
## $ Overall Survival Status    : chr [1:2509] "Living" "Living" "Deceased" "Living" ...
## $ PR Status                 : chr [1:2509] "Negative" "Positive" "Positive" "Positive" ...
## $ Radio Therapy             : chr [1:2509] "Yes" "Yes" "No" "Yes" ...
## $ Relapse Free Status (Months) : num [1:2509] 138.7 83.5 151.3 162.8 18.6 ...
## $ Relapse Free Status       : chr [1:2509] "Not Recurred" "Not Recurred" "Recurred" "Not Recurred" ...
## $ Sex                       : chr [1:2509] "Female" "Female" "Female" "Female" ...
## $ 3-Gene classifier subtype  : chr [1:2509] "ER-/HER2-" "ER+/HER2- High Prolif" NA NA ...
## $ Tumor Size                : num [1:2509] 22 10 15 25 40 31 10 65 29 34 ...
## $ Tumor Stage               : num [1:2509] 2 1 2 2 2 4 2 3 2 2 ...
## $ Patient's Vital Status     : chr [1:2509] "Living" "Living" "Died of Disease" "Living" ...
## - attr(*, "spec")=
## .. cols(
## ..   'Patient ID' = col_character(),
## ..   'Age at Diagnosis' = col_double(),
## ..   'Type of Breast Surgery' = col_character(),
## ..   'Cancer Type' = col_character(),
## ..   'Cancer Type Detailed' = col_character(),
## ..   Cellularity = col_character(),
## ..   Chemotherapy = col_character(),
## ..   'Pam50 + Claudin-low subtype' = col_character(),
## ..   Cohort = col_double(),
## ..   'ER status measured by IHC' = col_character(),
## ..   'ER Status' = col_character(),
## ..   'Neoplasm Histologic Grade' = col_double(),
## ..   'HER2 status measured by SNP6' = col_character(),
## ..   'HER2 Status' = col_character(),
## ..   'Tumor Other Histologic Subtype' = col_character(),
## ..   'Hormone Therapy' = col_character(),
## ..   'Inferred Menopausal State' = col_character(),
## ..   'Integrative Cluster' = col_character(),
## ..   'Primary Tumor Laterality' = col_character(),
## ..   'Lymph nodes examined positive' = col_double(),
## ..   'Mutation Count' = col_double(),
## ..   'Nottingham prognostic index' = col_double(),

```

```
## .. 'Uncotree Code' = col_character(),
## .. 'Overall Survival (Months)' = col_double(),
## .. 'Overall Survival Status' = col_character(),
## .. 'PR Status' = col_character(),
## .. 'Radio Therapy' = col_character(),
## .. 'Relapse Free Status (Months)' = col_double(),
## .. 'Relapse Free Status' = col_character(),
## .. Sex = col_character(),
## .. '3-Gene classifier subtype' = col_character(),
## .. 'Tumor Size' = col_double(),
## .. 'Tumor Stage' = col_double(),
## .. 'Patient's Vital Status' = col_character()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
# Quick summary of the dataset
summary(metabric)
```

```
## Patient ID      Age at Diagnosis Type of Breast Surgery Cancer Type
## Length:2509     Min.      :21.93      Length:2509      Length:2509
## Class :character 1st Qu.:50.92      Class :character  Class :character
## Mode  :character Median :61.11      Mode  :character  Mode  :character
##                               Mean  :60.42
##                               3rd Qu.:70.00
##                               Max.  :96.29
##                               NA's   :11
## Cancer Type Detailed Cellularity      Chemotherapy
## Length:2509      Length:2509      Length:2509
## Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character
##
##
##
## Pam50 + Claudin-low subtype      Cohort      ER status measured by IHC
## Length:2509      Min.      :1.0      Length:2509
## Class :character  1st Qu.:1.0      Class :character
## Mode  :character  Median :3.0      Mode  :character
##                               Mean  :2.9
##                               3rd Qu.:4.0
##                               Max.  :9.0
##                               NA's   :11
## ER Status      Neoplasm Histologic Grade HER2 status measured by SNP6
## Length:2509     Min.      :1.000      Length:2509
## Class :character 1st Qu.:2.000      Class :character
## Mode  :character Median :3.000      Mode  :character
##                               Mean  :2.412
##                               3rd Qu.:3.000
##                               Max.  :3.000
##                               NA's   :121
## HER2 Status      Tumor Other Histologic Subtype Hormone Therapy
## Length:2509      Length:2509      Length:2509
## Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character
```

```

##
##
##
##
## Inferred Menopausal State Integrative Cluster Primary Tumor Laterality
## Length:2509          Length:2509          Length:2509
## Class :character      Class :character      Class :character
## Mode  :character      Mode  :character      Mode  :character
##
##
##
##
## Lymph nodes examined positive Mutation Count    Nottingham prognostic index
## Min.   : 0.000          Min.   : 1.000          Min.   :1.000
## 1st Qu.: 0.000          1st Qu.: 3.000          1st Qu.:3.048
## Median : 0.000          Median : 5.000          Median :4.044
## Mean   : 1.951          Mean   : 5.579          Mean   :4.029
## 3rd Qu.: 2.000          3rd Qu.: 7.000          3rd Qu.:5.040
## Max.   :45.000          Max.   :80.000          Max.   :7.200
## NA's   :266             NA's   :152             NA's   :222
## Oncotree Code          Overall Survival (Months) Overall Survival Status
## Length:2509           Min.   : 0.00          Length:2509
## Class :character      1st Qu.: 60.87          Class :character
## Mode  :character      Median :116.47          Mode  :character
##                               Mean   :125.24
##                               3rd Qu.:185.13
##                               Max.   :355.20
##                               NA's   :528
## PR Status             Radio Therapy          Relapse Free Status (Months)
## Length:2509           Length:2509          Min.   : 0.00
## Class :character      Class :character      1st Qu.: 40.56
## Mode  :character      Mode  :character      Median : 99.09
##                               Mean   :108.84
##                               3rd Qu.:167.64
##                               Max.   :384.21
##                               NA's   :121
## Relapse Free Status    Sex                  3-Gene classifier subtype
## Length:2509           Length:2509          Length:2509
## Class :character      Class :character      Class :character
## Mode  :character      Mode  :character      Mode  :character
##
##
##
##
## Tumor Size            Tumor Stage          Patient's Vital Status
## Min.   : 1.00          Min.   :0.000          Length:2509
## 1st Qu.: 17.00          1st Qu.:1.000          Class :character
## Median : 22.41          Median :2.000          Mode  :character
## Mean   : 26.22          Mean   :1.714
## 3rd Qu.: 30.00          3rd Qu.:2.000
## Max.   :182.00          Max.   :4.000
## NA's   :149            NA's   :721

```

```

# Cleaning my dataset for analysis
metabric_clean <- metabric %>%
  rename(
    time_raw = `Overall Survival (Months)`,
    status_raw = `Overall Survival Status`,
    age = `Age at Diagnosis`,
    stage = `Tumor Stage`,
    size = `Tumor Size`,
    grade = `Neoplasm Histologic Grade`,
    ER = `ER Status`,
    PR = `PR Status`,
    HER2 = `HER2 Status`
  ) %>%
  filter(
    !is.na(time_raw),
    !is.na(status_raw),
    !is.na(stage),
    time_raw > 0
  ) %>%
  mutate(
    time = time_raw,
    event = ifelse(status_raw == "Deceased", 1, 0),
    stage = factor(stage),
    grade = factor(grade),
    ER = factor(ER),
    PR = factor(PR),
    HER2 = factor(HER2)
  ) %>%
  select(time, event, age, stage, size, grade, ER, PR, HER2)

dim(metabric_clean)

```

```
## [1] 1466    9
```

```
table(metabric_clean$event)
```

```
##
##    0    1
## 643 823
```

```
table(metabric_clean$stage, useNA = "always")
```

```
##
##    0    1    2    3    4 <NA>
##   12  501  825  118   10     0
```

```

# Overall Kaplan-Meier
km_overall <- survfit(Surv(time, event) ~ 1, data = metabric_clean)

ggsurvplot(
  km_overall,

```

```

data = metabric_clean,
conf.int = TRUE,
risk.table = TRUE,
risk.table.y.text = FALSE,
risk.table.height = 0.25,
tables.theme = theme_cleantable(),
xlab = "Time (Months)",
ylab = "Survival Probability",
title = "Kaplan-Meier Overall Survival",
ggtheme = theme_bw(),
tables.ggtheme = theme_bw()
)

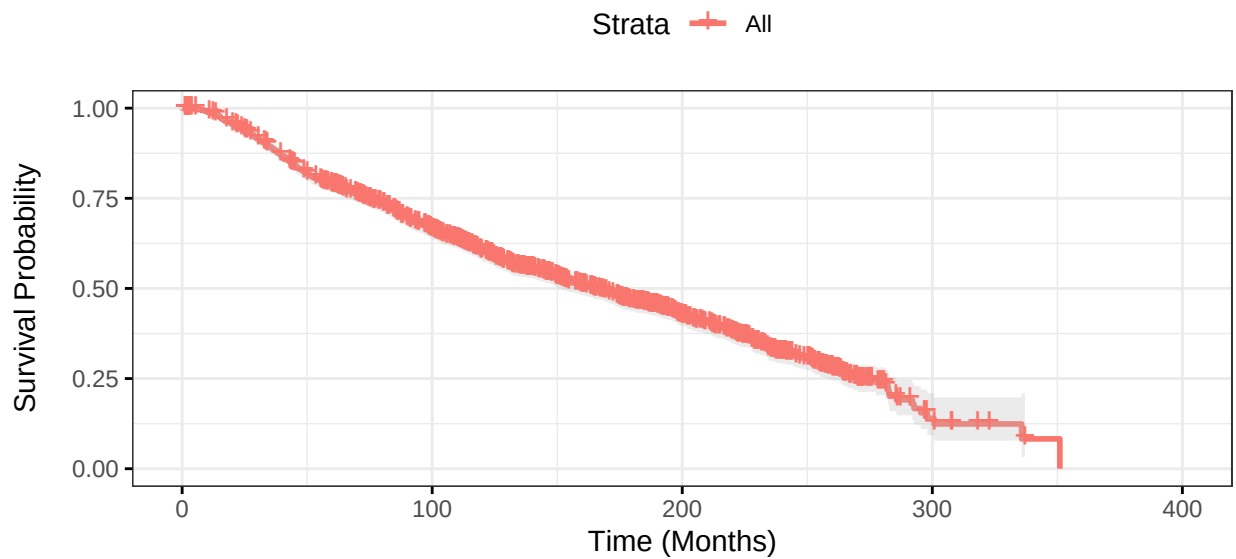
```

```

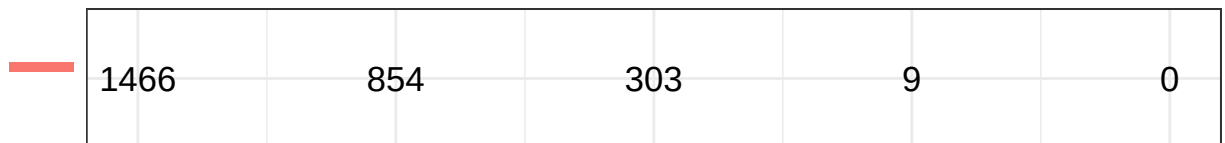
## Ignoring unknown labels:
## * fill : "Strata"
## Ignoring unknown labels:
## * fill : "Strata"
## Ignoring unknown labels:
## * fill : "Strata"
## Ignoring unknown labels:
## * fill : "Strata"
## Ignoring unknown labels:
## * colour : "Strata"

```

Kaplan-Meier Overall Survival



Number at risk



```
# Stage-specific Kaplan-Meier
```

```
metabric_stage <- metabric_clean %>%  
  filter(!is.na(stage), stage != "0") %>%  
  mutate(stage = droplevels(stage))
```

```
table(metabric_stage$stage)
```

```
##
```

```
##    1    2    3    4
```

```
## 501 825 118  10
```

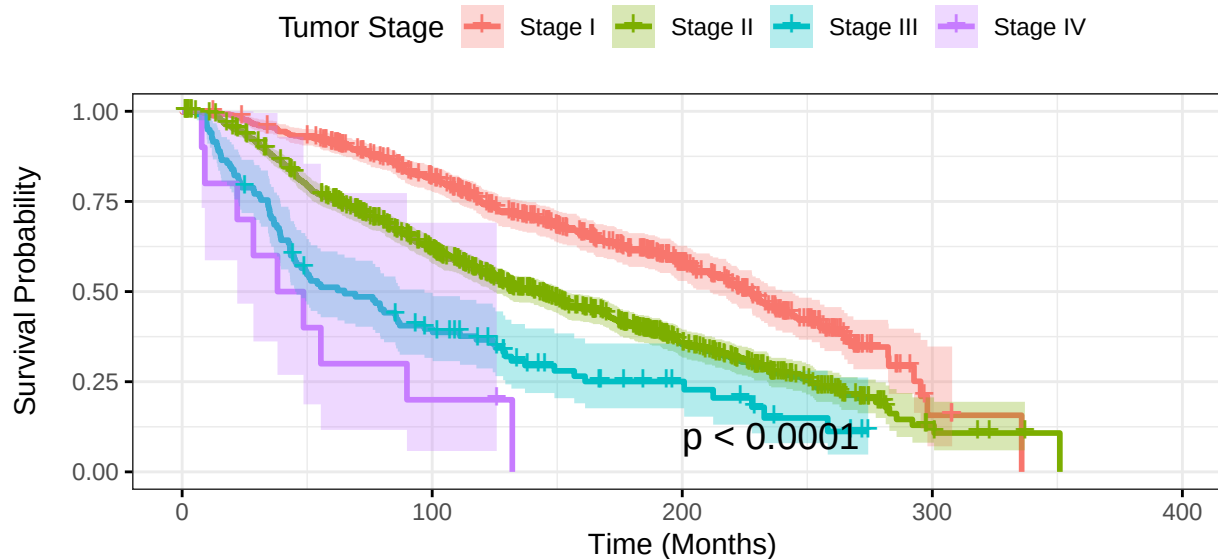
```
km_stage <- survfit(Surv(time, event) ~ stage, data = metabric_stage)
```

```
ggsurvplot(km_stage,  
  data = metabric_stage,  
  conf.int = TRUE,  
  risk.table = TRUE,  
  risk.table.y.text = FALSE,  
  risk.table.height = 0.25,  
  tables.theme = theme_cleantable(),  
  pval = TRUE,  
  pval.coord = c(200, 0.1),  
  legend.title = "Tumor Stage",  
  legend.labs = c("Stage I", "Stage II", "Stage III", "Stage IV"),  
  xlab = "Time (Months)",  
  ylab = "Survival Probability",  
  title = "Kaplan-Meier Survival by Tumor Stage",  
  ggtheme = theme_bw(),  
  tables.ggtheme = theme_bw())
```

```
## Ignoring unknown labels:
```

```
## * colour : "Tumor Stage"
```

Kaplan–Meier Survival by Tumor Stage



Number at risk

Stage I	501	363	156	3	0
Stage II	825	442	135	6	0
Stage III	118	42	11	0	0
Stage IV	10	2	0	0	0

Running a Cox PH model with key clinical predictors

```
cox_model <- coxph(Surv(time, event) ~ age + stage + size + grade + ER + PR + HER2,
  data = metabric_clean)
```

```
summary(cox_model)
```

Call:

```
## coxph(formula = Surv(time, event) ~ age + stage + size + grade +
##       ER + PR + HER2, data = metabric_clean)
```

##

```
## n= 1406, number of events= 785
```

```
## (60 observations deleted due to missingness)
```

##

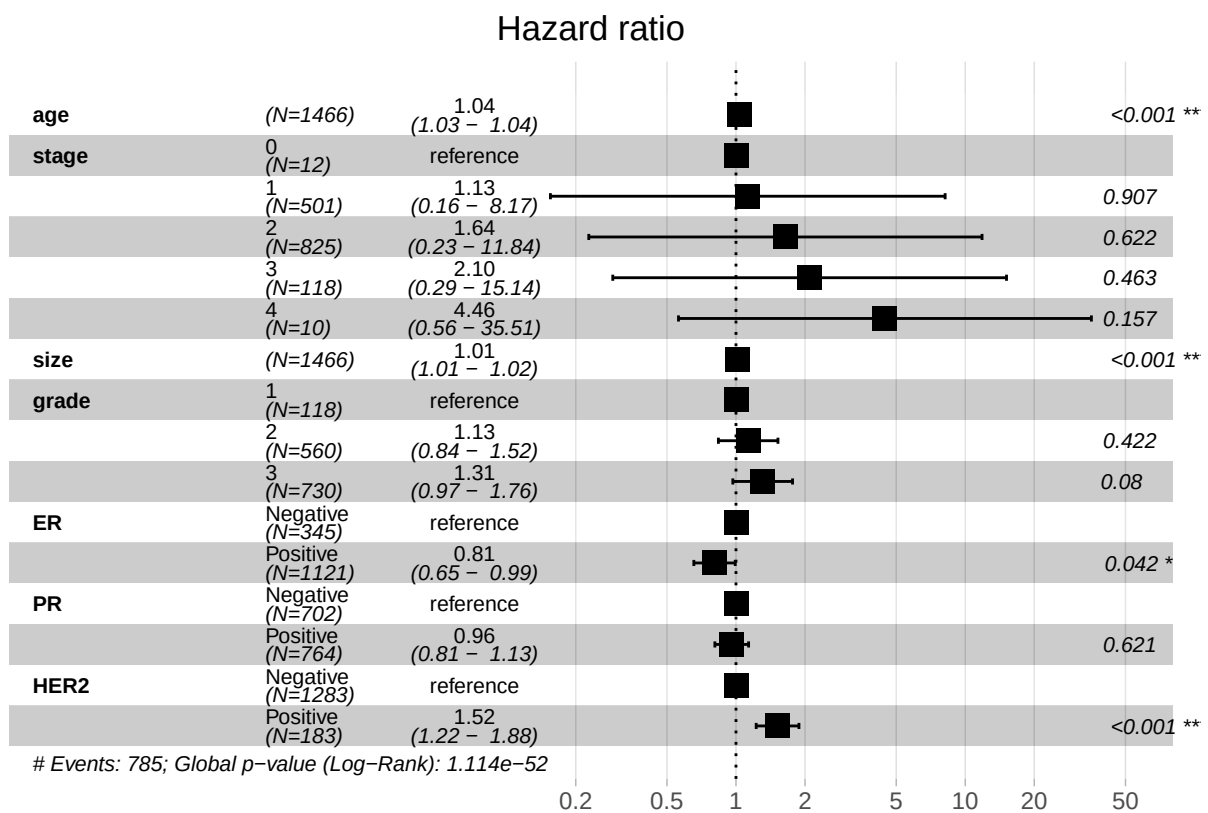
```
##               coef exp(coef) se(coef)      z Pr(>|z|)
## age           0.035729  1.036374  0.003241 11.025 < 2e-16 ***
## stage1        0.118364  1.125654  1.011424  0.117  0.90684
## stage2        0.496339  1.642696  1.007638  0.493  0.62231
## stage3        0.739860  2.095643  1.008903  0.733  0.46336
## stage4        1.496073  4.464122  1.058044  1.414  0.15736
## size          0.011135  1.011198  0.002384  4.671   3e-06 ***
## grade2        0.122412  1.130220  0.152557  0.802  0.42232
## grade3        0.268062  1.307428  0.152967  1.752  0.07970 .
## ERPositive    -0.215569  0.806083  0.106040 -2.033  0.04206 *
## PRPositive    -0.042699  0.958200  0.086377 -0.494  0.62107
## HER2Positive  0.417425  1.518047  0.109593  3.809  0.00014 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## age              1.0364    0.9649    1.0298    1.0430
## stage1           1.1257    0.8884    0.1551    8.1720
## stage2           1.6427    0.6088    0.2280   11.8375
## stage3           2.0956    0.4772    0.2901   15.1390
## stage4           4.4641    0.2240    0.5612   35.5095
## size             1.0112    0.9889    1.0065    1.0159
## grade2           1.1302    0.8848    0.8381    1.5241
## grade3           1.3074    0.7649    0.9688    1.7645
## ERPositive       0.8061    1.2406    0.6548    0.9923
## PRPositive       0.9582    1.0436    0.8090    1.1350
## HER2Positive     1.5180    0.6587    1.2246    1.8818
##
## Concordance= 0.664  (se = 0.011 )
## Likelihood ratio test= 275.7  on 11 df,   p=<2e-16
## Wald test               = 282.8  on 11 df,   p=<2e-16
## Score (logrank) test = 297.9  on 11 df,   p=<2e-16
```

```
# Visualizing the Cox model results
metabric_clean_df <- as.data.frame(metabric_clean)

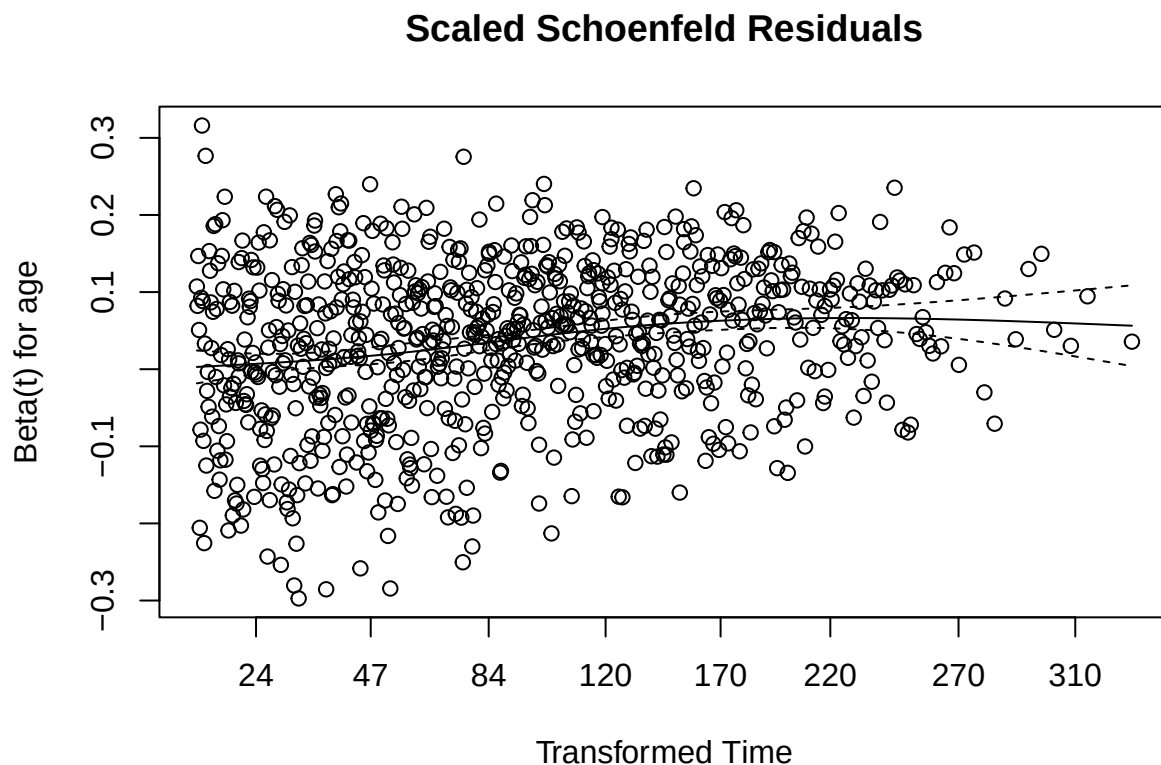
ggforest(cox_model, data = metabric_clean_df)
```



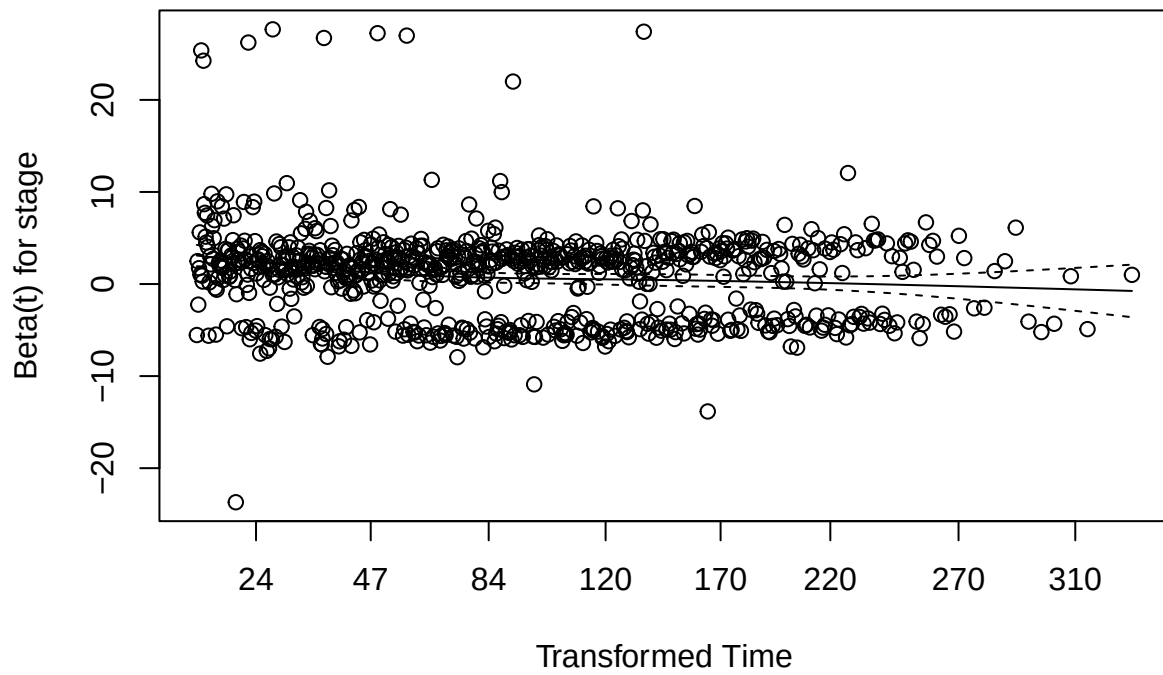
```
# Testing proportional hazards assumption
cox_zph_test <- cox.zph(cox_model)
print(cox_zph_test) # Smaller p-values suggest evidence against PH
```

```
##          chisq df      p
## age      67.262  1 2.4e-16
## stage    25.386  4 4.2e-05
## size      0.995  1 0.31861
## grade     45.009  2 1.7e-10
## ER        61.320  1 4.9e-15
## PR        30.504  1 3.3e-08
## HER2      11.544  1 0.00068
## GLOBAL   140.783 11 < 2e-16
```

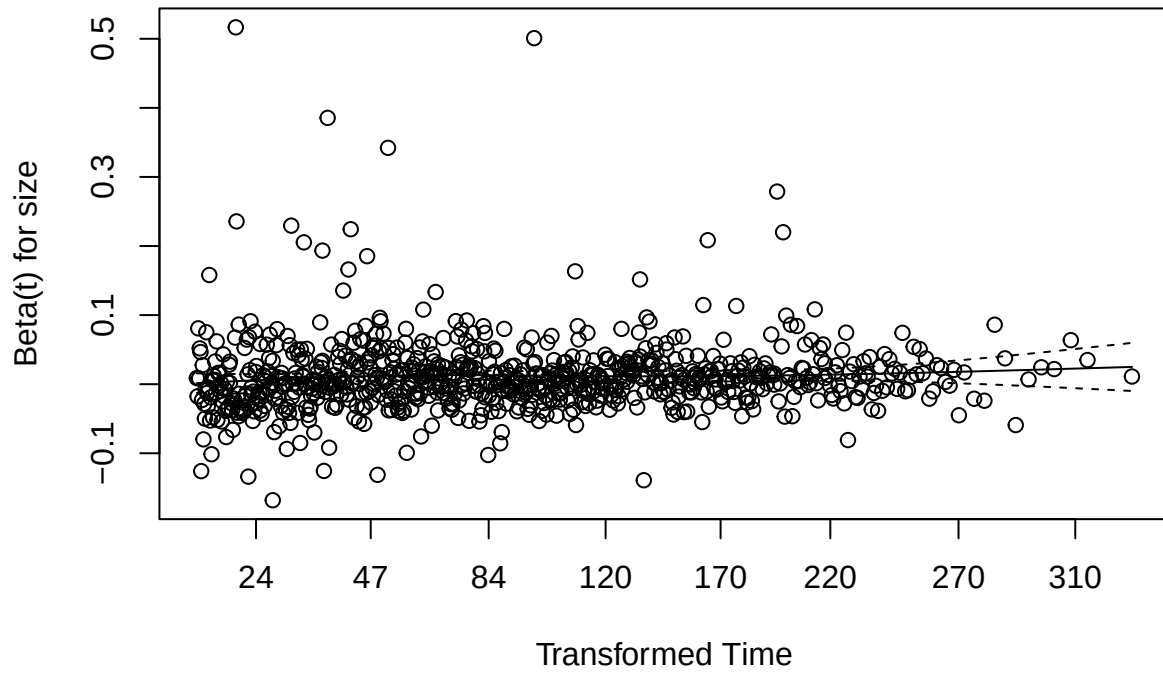
```
# Plotting residuals (straight lines = assumption holds)
plot(cox_zph_test,
     main = "Scaled Schoenfeld Residuals",
     xlab = "Transformed Time")
```



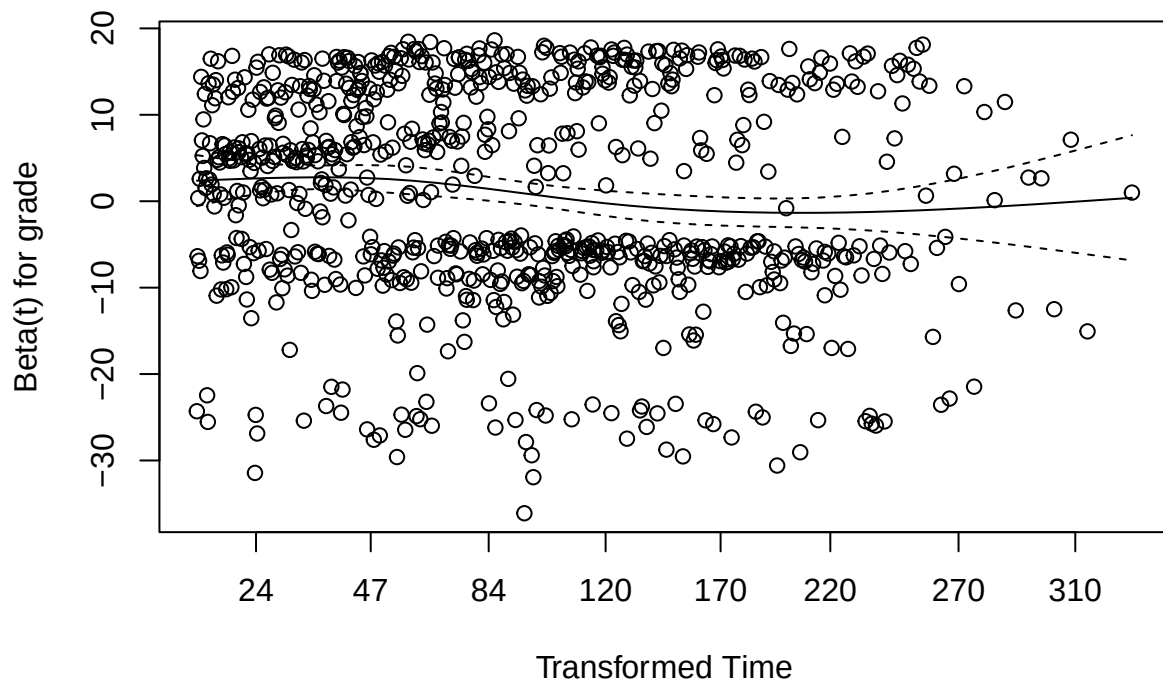
Scaled Schoenfeld Residuals



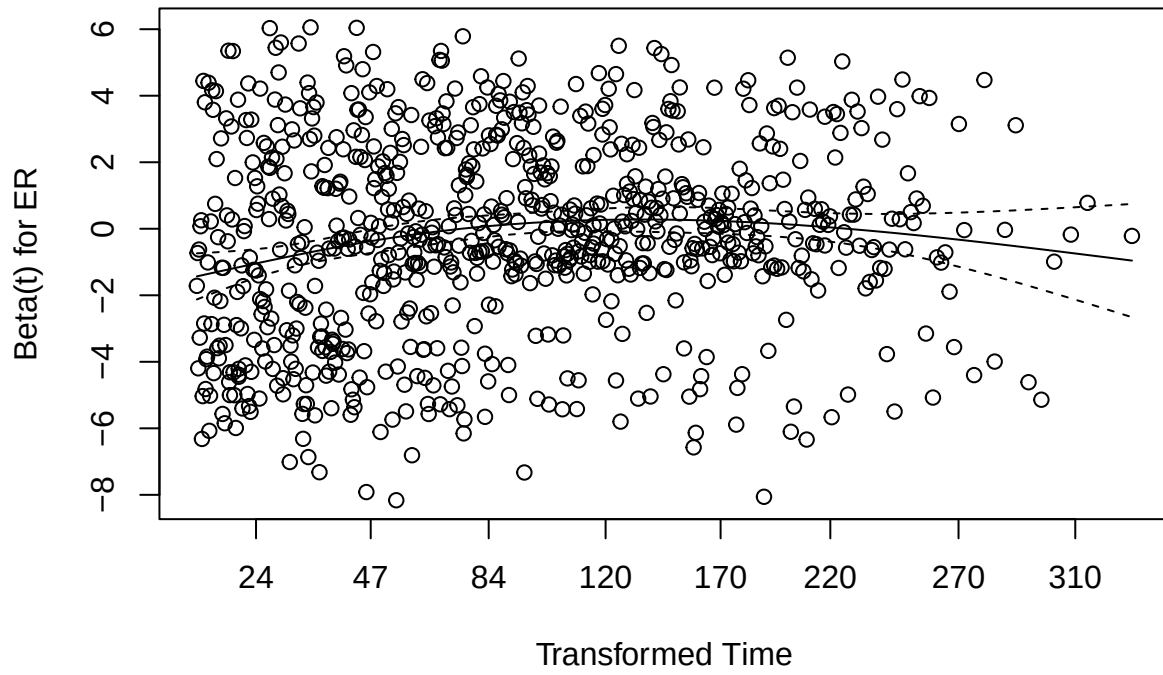
Scaled Schoenfeld Residuals



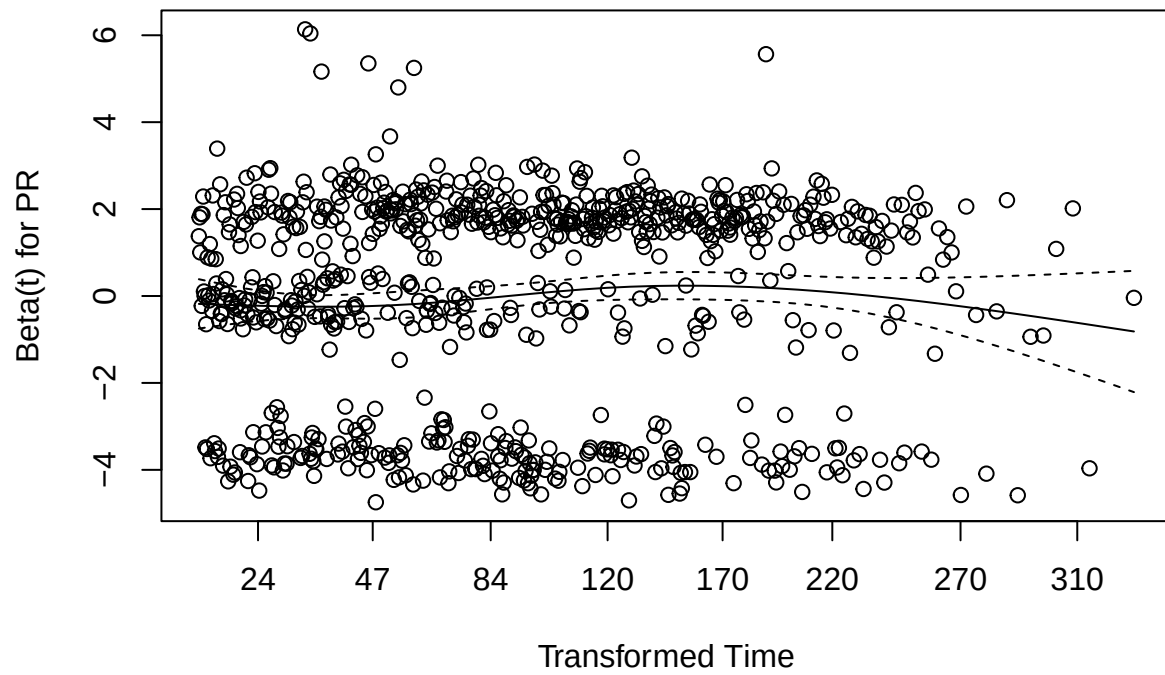
Scaled Schoenfeld Residuals



Scaled Schoenfeld Residuals

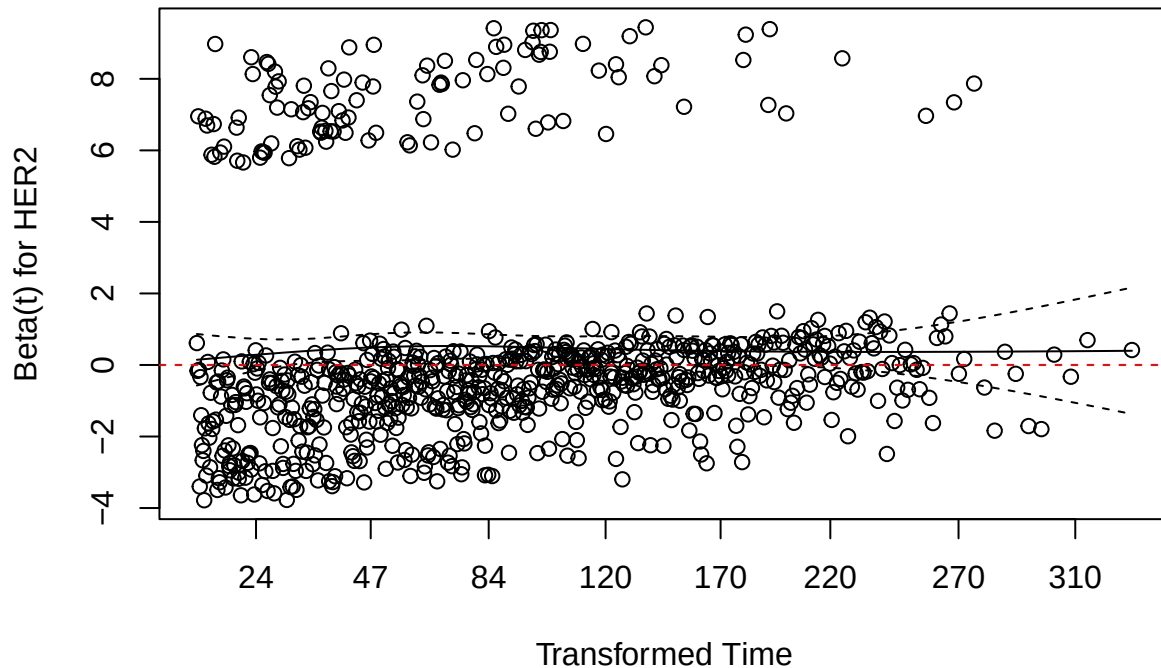


Scaled Schoenfeld Residuals



```
abline(h = 0, col = "red", lty = 2)
```

Scaled Schoenfeld Residuals



```
# Revised Cox model to address PH violation and the following was done:
# - stage was handled with stratification
# - age was handled as a time-dependent covariate
cox_revised <- coxph(
  Surv(time, event) ~ age + tt(age) + size + grade + ER + PR + HER2 + strata(stage),
  data = metabriz_clean,
  tt = function(x, t, ...) x * log(t + 1)
)

summary(cox_revised)
```

```
## Call:
## coxph(formula = Surv(time, event) ~ age + tt(age) + size + grade +
##       ER + PR + HER2 + strata(stage), data = metabriz_clean, tt = function(x,
##       t, ...) x * log(t + 1))
##
##      n= 1406, number of events= 785
##      (60 observations deleted due to missingness)
##
##              coef exp(coef)  se(coef)      z Pr(>|z|)
## age          -0.082142  0.921142  0.015776 -5.207 1.92e-07 ***
## tt(age)         0.028057  1.028454  0.003723  7.536 4.84e-14 ***
## size           0.010188  1.010240  0.002427  4.197 2.70e-05 ***
## grade2         0.123170  1.131077  0.152620  0.807 0.419645
## grade3         0.259656  1.296483  0.153143  1.696 0.089980 .
## ERPositive    -0.211206  0.809608  0.106224 -1.988 0.046777 *
```

```
## PRPositive    -0.060093  0.941677  0.086478 -0.695 0.487118
## HER2Positive  0.390645  1.477934  0.110416  3.538 0.000403 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## age              0.9211    1.0856    0.8931    0.9501
## tt(age)          1.0285    0.9723    1.0210    1.0360
## size             1.0102    0.9899    1.0054    1.0151
## grade2           1.1311    0.8841    0.8387    1.5255
## grade3           1.2965    0.7713    0.9603    1.7503
## ERPositive       0.8096    1.2352    0.6574    0.9970
## PRPositive       0.9417    1.0619    0.7949    1.1156
## HER2Positive     1.4779    0.6766    1.1903    1.8350
##
## Concordance= 0.647 (se = 0.012 )
## Likelihood ratio test= 220.6 on 8 df,  p=<2e-16
## Wald test              = 208.6 on 8 df,  p=<2e-16
## Score (logrank) test = 216.7 on 8 df,  p=<2e-16
```

```
# Note:
# cox.zph() is not used on cox_revised because models with tt() terms
# are not compatible with the standard Schoenfeld residual test
```